



UNIVERSITY OF EDINBURGH
Business School

2025 - 26

(CMSE11432) Principles of Data Analytics

**The Multiplier Effect on International Graduate Career
Success**

Workshop 3 – Group 1

Word count: 1489

1. Background, Significance, and Research Questions

For people graduating internationally, navigating through job market is very challenging, often compounded by cultural adjustments, visa constraints, and the need to validate foreign credentials in a local context (Crossman & Clarke, 2010). Moving forward with Human Capital Theory, it assumes that skills acquired through education and experience are investments that produce economic returns (Becker, 1964). Practical experience like internships (Knouse & Fontenot, 2008), education level (Hout, 2012) and host-country language proficiency (Chiswick & Miller, 2007) are the key factors of job market outcomes.

This report presents a rigorous data analysis case study to showcase the application of quantitative methods to investigate the determinants of career success for international graduates. Our analysis aims to provide a data-driven strategy to guide career planning. Therefore, this study moves beyond simple correlations to quantify the combined effects of skills and qualifications on measurable career outcomes: employment status and annual salary. The data are analysed using descriptive statistics, hypothesis testing and ANOVA.

This report focuses on two main perspectives

1. Evaluate the effect of international graduates' language proficiency, educational level and internship experience, both individually and combined, on employment status
2. Further evaluate the significance of these variables on annual salary after international graduates were employed

2. Data Source and Presentation

The data for this case study is a sample of 300,000 international graduates from the *International Graduates Employment Dataset* on Kaggle. This dataset contains variables on demographics, academic background, and employment outcomes which was sourced from UK HESA Graduate Outcomes Survey and Eurostat EU Labour Force Survey (EU-LFS). A total of 143,356 missing values (47.78%) were identified in the Job Sector variable and were imputed with the category 'Unknown' to retain the full sample.

2.1 Key Variables

1. **Independent Variable (Categorical):** Language proficiency (Basic, Intermediate, Advanced, Fluent), Education Level (Diploma, Bachelor's, Master's, PhD), Internship Experience (Yes, No)
2. **Dependent Variable (Categorical):** Employment Status (Employed, unemployed, continuing education).
3. **Dependent Variable (Continuous):** Salary (annual salary, filtered only for 'Employed' graduates).

2.2 Overview

Our sample is predominantly composed of bachelor's and master's degree holders, making up 75% of the total. Most respondents have internship experience around 65% and their language proficiency is mainly at the intermediate or advanced level. Over half of this group is already employed (see Figure 1), and the salary distribution presents multiple concentration points (see Figure 2).

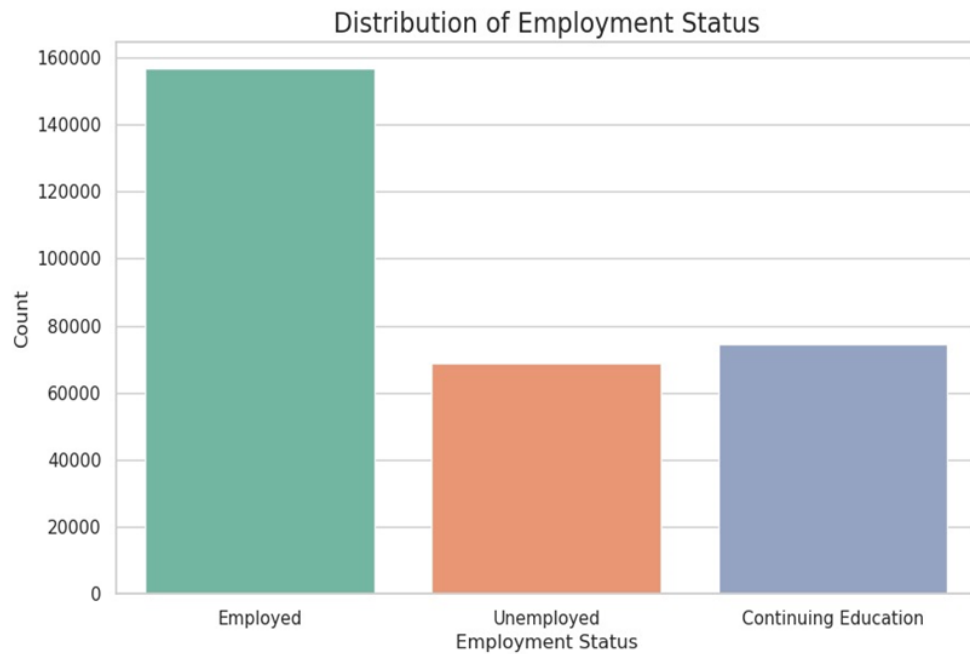


Figure 1. Distribution of Employment Status

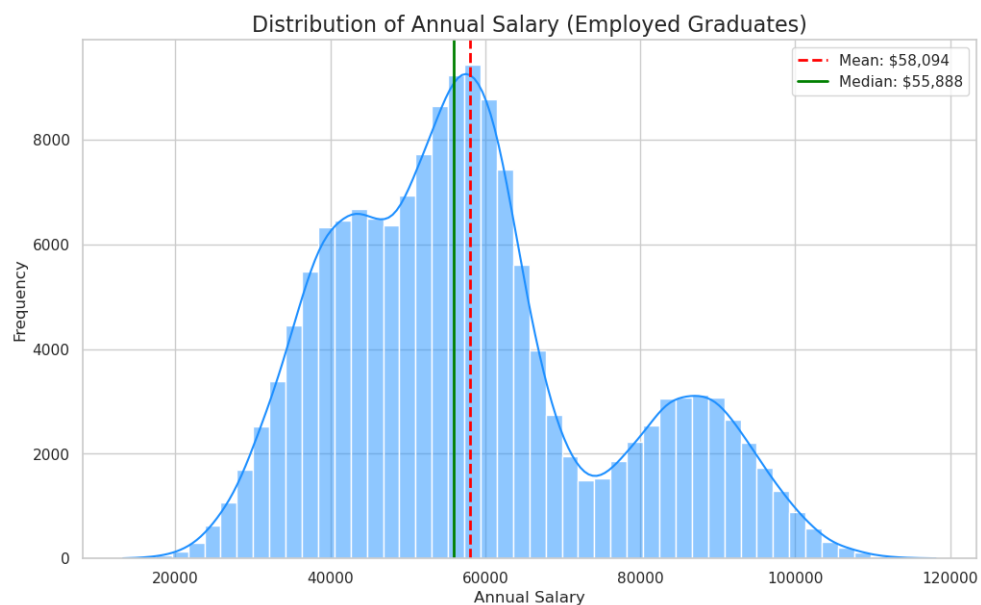


Figure 2. Distribution of Annual Salary

3. Methodology, Concepts, and Approaches

3.1 Employment Status (Categorical Analysis)

Contingency tables are first constructed to visualise frequency distributions on all combinations of independent and dependent variables (Agresti, 2018). Furthermore, Chi-square (χ^2) tests of independence will be conducted to determine the significance of statistical association between the relationship of the three independent variables and the dependent variable of employment status. (Heumann, Schomaker and Shalabh, 2016).

H_0 : Employment status and language proficiency/educational level/internship experience are independent

H_1 : There is a significant association between employment status and language proficiency/educational level/internship

Cramer's V is then calculated to measure the strength of statistical association based on the Chi-square statistic χ^2 value, a standardised measure of effect size from 0 (i.e. no association) to 1 (i.e. perfect association). Small ($V \approx 0.1$), medium ($V \approx 0.3$), and strong ($V \approx 0.5$) associations are identified (Cohen, 1988).

To evaluate possible interaction effects between independent variables, both individual variables (e.g. education level $\rightarrow$ employment status) and two combined variables (e.g., language proficiency $\times$ education level $\rightarrow$ employment status) are tested with the above statistical methods (Hair et al., 2019).

3.2 Salary (Continuous Analysis)

The dataset will be filtered for 'Employed' graduates only (which have Salary > 0) to ensure the validity of analysis. To explore salary differences, One-Way and Two-Way Analysis of Variance (ANOVA) were used. ANOVA is used to compare the mean salary across multiple groups (e.g. Education Level) and to test for interaction effects (e.g. Education Level $\times$ Internship Experience) (Field, 2018). F-statistic is calculated to decide the rejection of the null hypothesis that the means are similar across multiple groups. (Kumar, 2024)

H_0 : Mean salary is the same among all groups

H_1 : Mean salary is different among all groups

When ANOVA indicates significant differences between group means, Tukey-Kramer procedure will be conducted to identify which specific groups were significantly different from one another.

4. Analysis, Key Findings, and Discussion from the outcomes

4.1 Descriptive analysis

The figure below shows that the employment rate of graduates with internship experience (57%) is significantly higher than that of those without (42%). The latter are mostly in a state of "unemployed" or "continuing education," so we conclude that having internship experience exerts a positive impact on employment. Subsequent Chi-Square tests will quantify the strength of this relationship and compare it with other factors such as educational background and language proficiency.

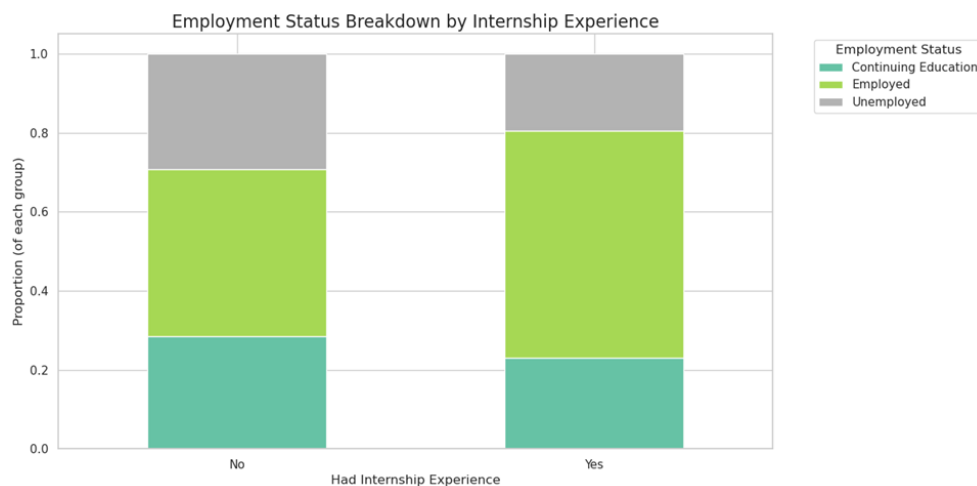


Figure 3. Employment Status Breakdown by Internship Experience

Figure 4 illustrates the combined impact of internships and education on salary. The median salary of graduates with internship experience is higher than that of those without, across all academic degree levels. However, our subsequent ANOVA analysis revealed that the relative value of this increase varies; specifically, internships have a more significant positive effect on the salaries of bachelor's degree graduates than on those of PhD graduates.

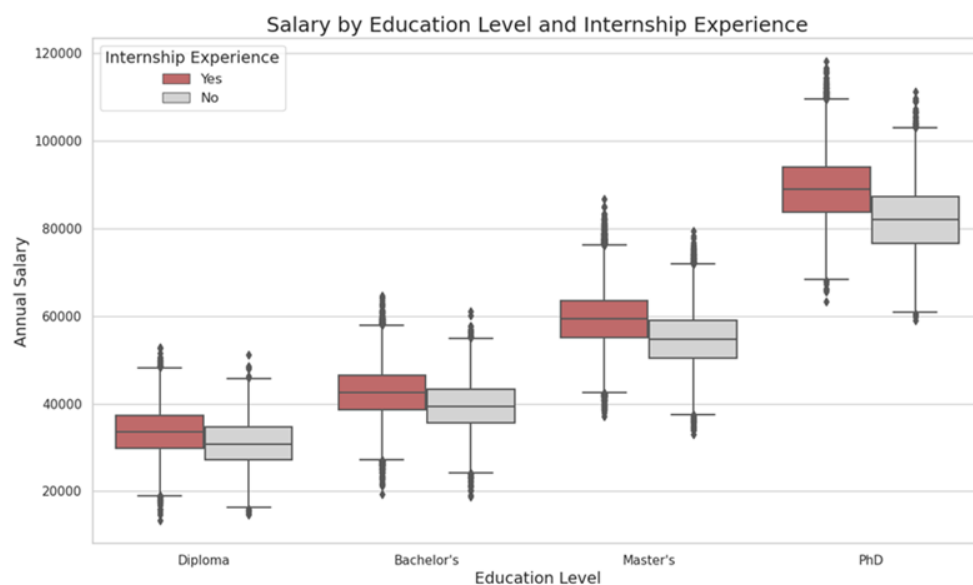


Figure 4. Salary by Education Level and Internship Experience

4.2 Employment status (Categorical Outcome)

All individual and combined independent variable tests were highly statistically significant ($p < 0.001$), indicating that all factors are related to Employment Status.

Independent Variables	Type	Chi2	df	p-value
Language_Proficiency × Education_Level	Combined	95948.64	30	0.000
Language_Proficiency × Internship_Experience	Combined	54611.94	14	0.000
Education_Level × Internship_Experience	Combined	53467.8	14	0.000
Language_Proficiency	Individual	47438.76	6	0.000
Education_Level	Individual	46124.19	6	0.000
Internship_Experience	Individual	6813.133	2	0.000

Table 1. Summary table of Chi-square test results and Cramér's V values

The findings confirm that Employment Status is multifaceted. As per the Cramér's V values (See Figure 5), Language Proficiency ($V=0.281$) and Education Level ($V=0.277$) show medium but the strongest association among all three independent variables individually, while Internship ($V=0.151$) has a small association with Employment Status.

Moreover, the key finding is the strong interaction effect. The combination of Language × Education ($V=0.400$) is the strongest predictor than any factor alone or combined. The returns to higher education are highest when combined with strong and effective language skills. The result provides strong evidence that universities should emphasise both academic development (human capital) and linguistic competencies enhancement (cultural capital) within their curricula to maximise international graduates' employability and career success (Becker, 1964).

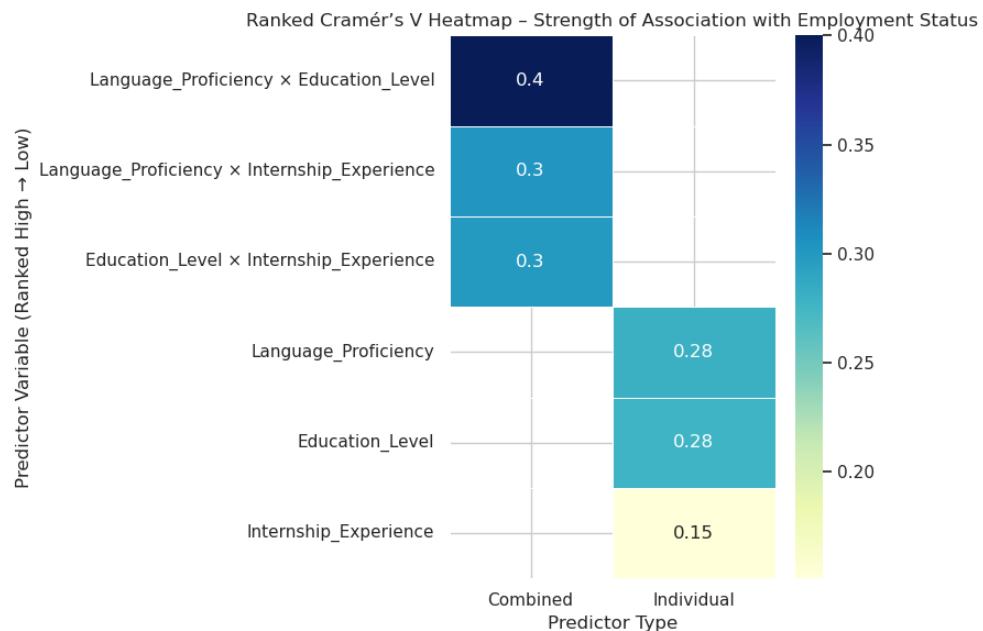


Figure 5. Ranked Cramér's V Heatmap

4.3 Salary (Continuous Outcome)

Analysis for salary was conducted using ANOVA and Tukey's tests on the subset of 156,644 employed graduates.

The One-Way ANOVA for Education Level was statistically significant, implying that salary differs significantly across different education levels. (F scores results = 48634.4, $p < 0.001$)

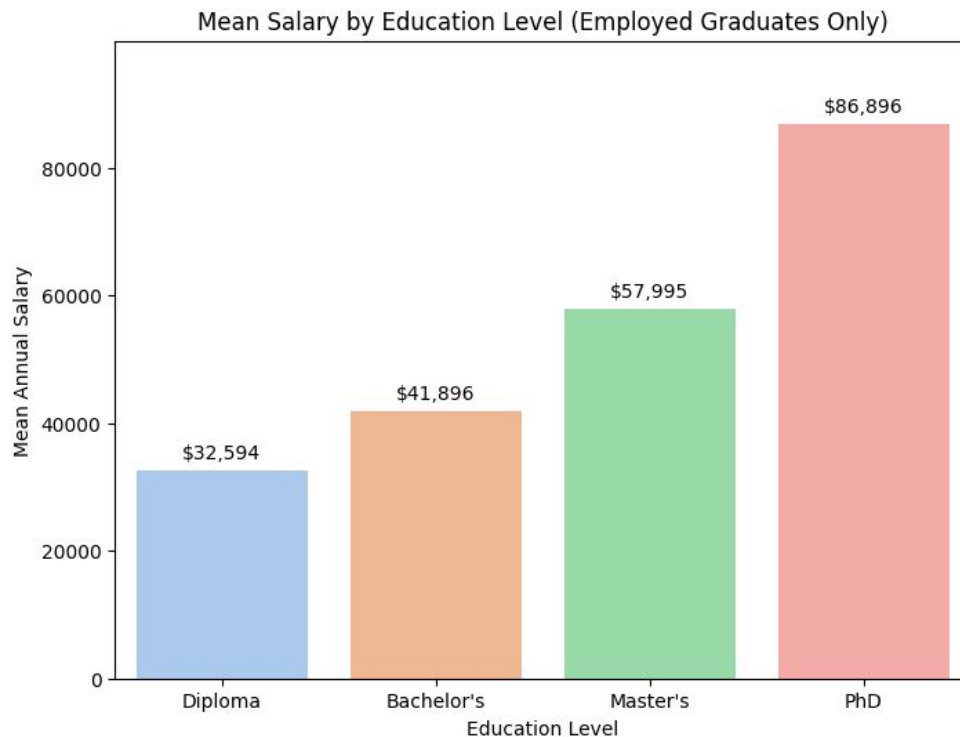


Figure 6. Mean Salary by Education Level (Employed Graduates Only)

The Tukey test also confirms a clear hierarchy: PhD > Master's > Bachelor's > Diploma.

Multiple Comparison of Means - Tukey HSD, FWER=0.05

group1	group2	meandiff	p-adj	lower	upper	reject
Bachelor's	Diploma	-3159.5064	0.0	-3592.0258	-2726.9869	True
Bachelor's	Master's	27098.0765	0.0	26815.082	27381.0711	True
Bachelor's	PhD	47509.6928	0.0	47139.0218	47880.3638	True
Diploma	Master's	30257.5829	0.0	29818.8907	30696.2751	True
Diploma	PhD	50669.1992	0.0	50169.4373	51168.9611	True
Master's	PhD	20411.6163	0.0	20033.7607	20789.4718	True

Figure 7. Tukey test for Education Level

The One Way ANOVA for Language Proficiency was also very much significant. (F scores results = 11770.5, $p < 0.001$)

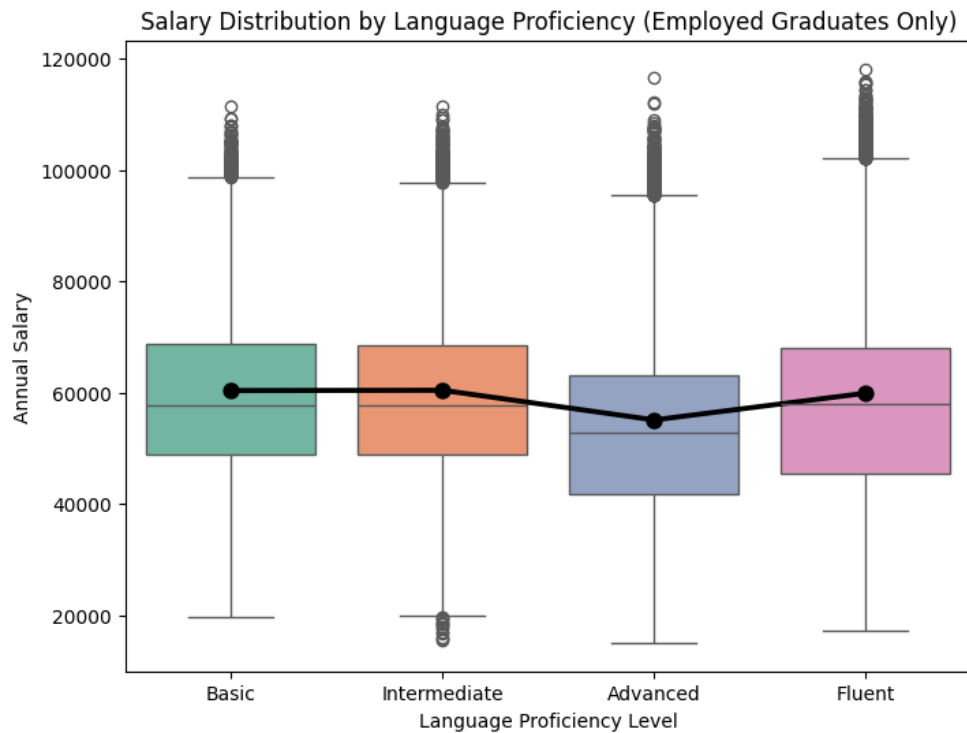


Figure 8. Salary Distribution by Language Proficiency (Employed Graduates Only)

The Two Way ANOVA for Education Level and Internship Experience also showed a significant interaction effect. (F scores results = 433.46, $p < 0.001$).

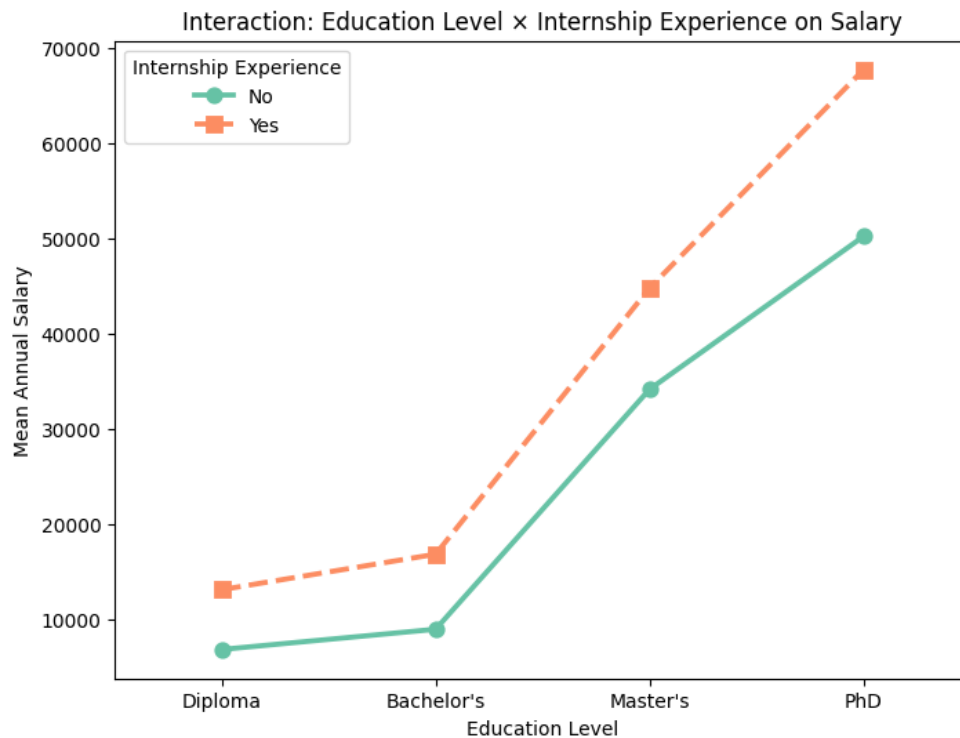


Figure 9. Mean Salary by Education Level x Internship Experience

The Two Way ANOVA for Education Level and Language Proficiency has also displayed significant interaction effect. (F scores results = 690.79, $p < 0.001$).

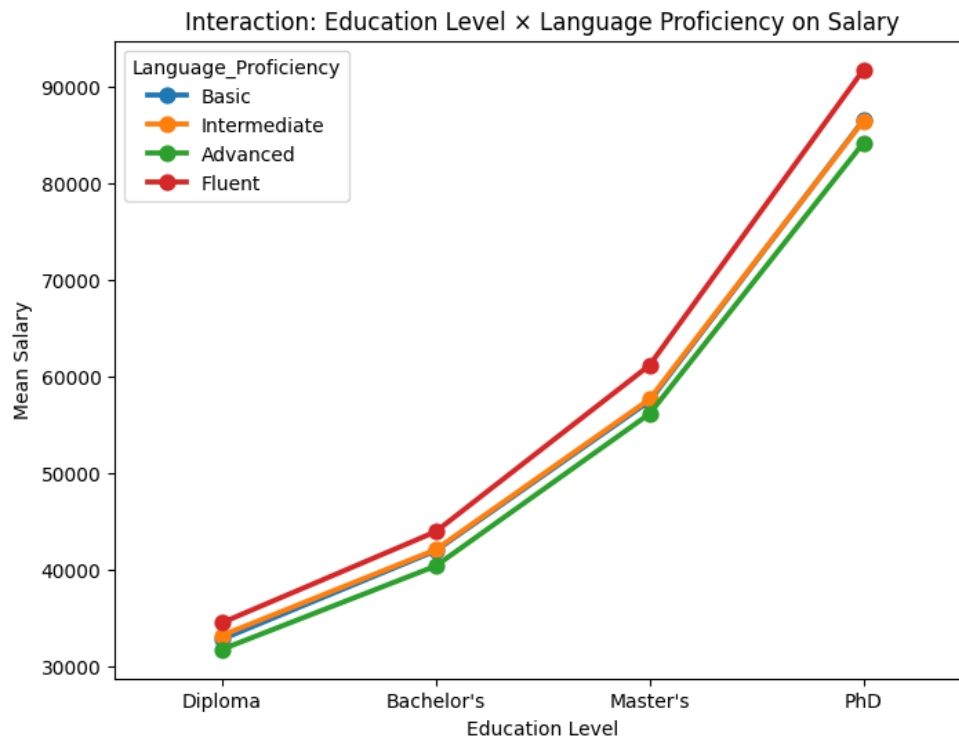


Figure 10. Mean Salary by Education Level x Language Proficiency

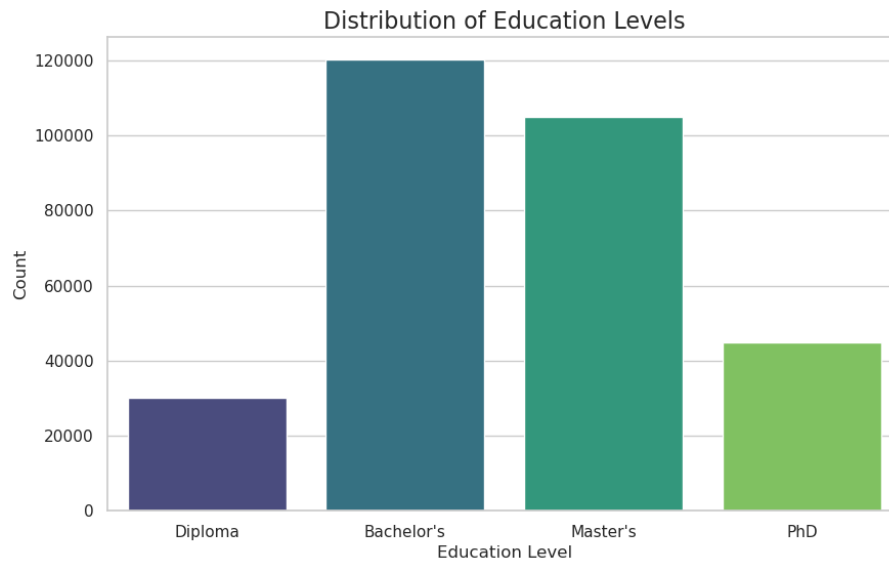
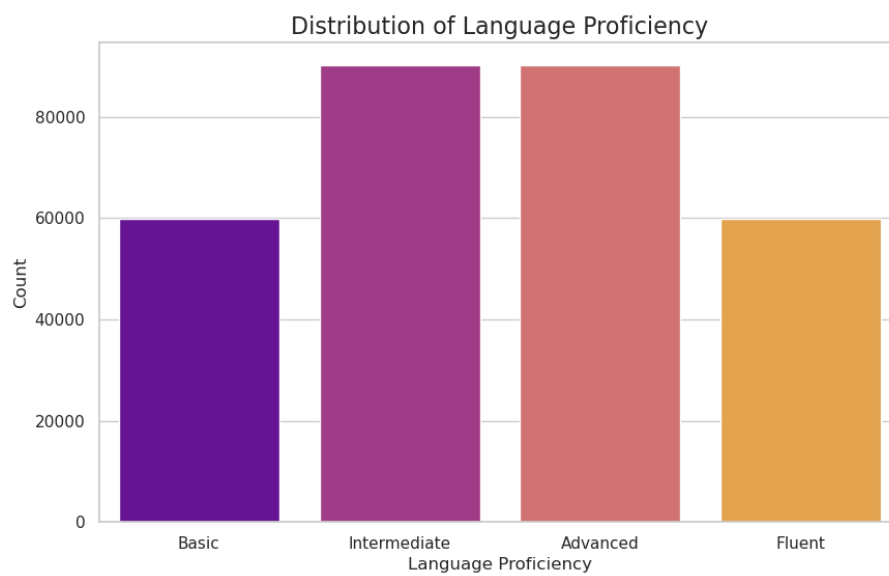
The significant interaction effect from the Education × Internship ANOVA (See Figure 9) is a key finding. It demonstrates that the salary benefit of an internship is not uniform across all education levels. As the chart illustrates, an internship provides a substantial salary boost for Bachelor's graduates but a much smaller relative boost for PhDs. This supports Human Capital Theory [Becker, 1964]. For a Bachelor's graduate, an internship signals practical readiness (human capital) that their degree alone may not, thus boosting their market value. A PhD graduate's degree, however, already signals high-level specialised expertise; their salary is primarily determined by this specialisation, making the generalist signal of an internship less impactful on salary. The second interaction (See Figure 10) between Education and Language further reinforces this. The salary gap between 'Basic' and 'Advanced/Fluent' language skills widens significantly at the Master's and PhD levels. This suggests that for high-level roles, advanced technical knowledge (the degree) must be paired with advanced communication skills (language) to unlock the highest salary potential.

5. Limitations and Further Research

A key limitation was the 'Unknown' Job Sector for 143,356 graduates, which prevented a detailed analysis of salary differences across industries. Additionally, a multinomial logistic regression model could be developed to estimate the probability of graduates falling into one of the three Employment Status categories, providing a more predictive approach than the chi-square test of independence alone.

References

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Appendix A*Figure A-1 Distribution of Education Levels**Figure A-2 Distribution of Language Proficiency*

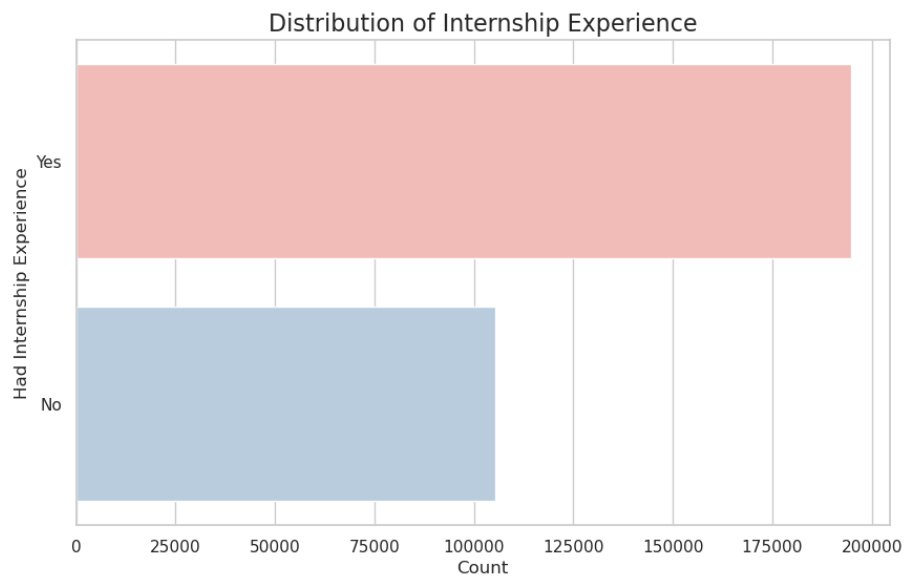


Figure A-3 Distribution of Internship Experience

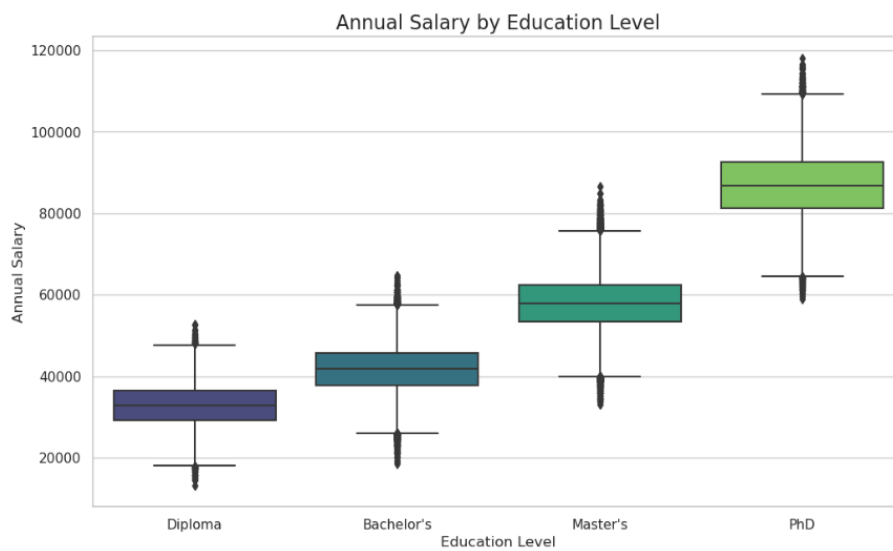


Figure A-4 Annual Salary by Education Level

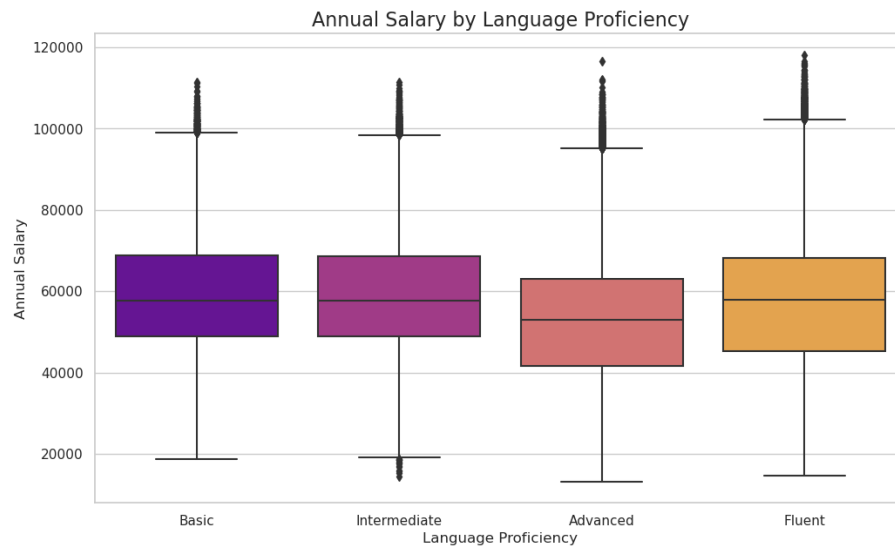


Figure A-5 Annual Salary by Language Proficiency

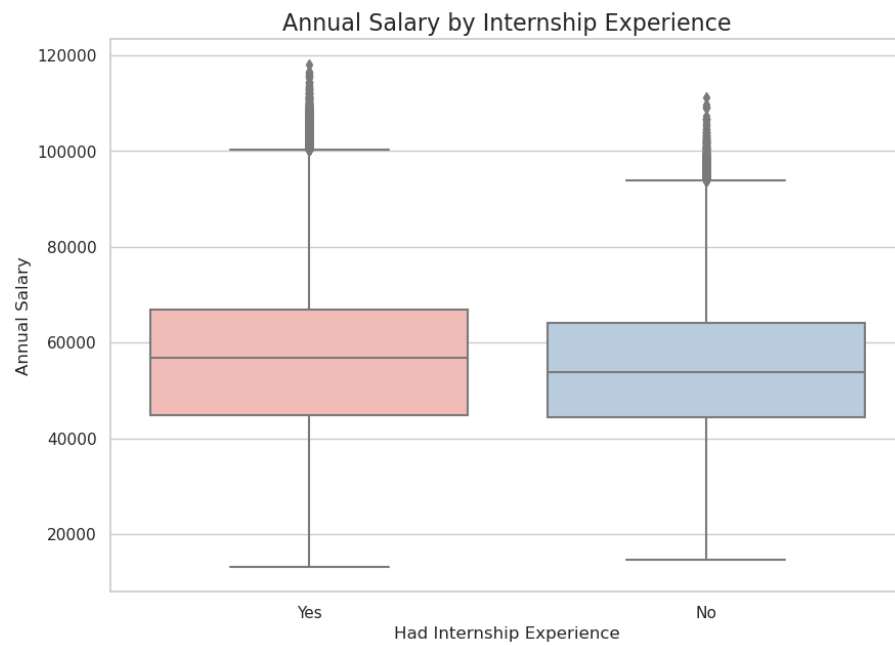


Figure A-6 Annual Salary by Internship Experience

Appendix B**Employment Status**

Language Proficiency	Continuing Education	Employed	Unemployed	All
Basic	18469	19952	21460	59881
Intermediate	27648	29887	32598	90133
Advanced	16969	64281	8861	90111
Fluent	11531	42524	5820	59875
All	74617	156644	68739	300000

*Table B-1 Contingency table of Language Proficiency and Employment Status***Employment Status**

Education Level	Continuing Education	Employed	Unemployed	All
Diploma	9234	9939	10802	29975
Bachelor's	36714	40377	43052	120143
Master's	20053	74573	10397	105023
PhD	8616	31755	4488	44859
All	74617	156644	68739	300000

*Table B-2 Contingency table of Education Level and Employment Status***Employment Status**

Internship Experience	Continuing Education	Employed	Unemployed	All
Yes	44705	112150	37855	194710
No	29912	44494	30884	105290
All	74617	156644	68739	300000

Table B-3 Contingency table of Internship Experience and Employment Status

Language Proficiency × Education Level	Employment Status			
	Continuing Education	Employed	Unemployed	All
Basic & Diploma	2018	855	3148	6021
Basic & Bachelor's	8045	3618	12395	24058
Basic & Master's	5851	10822	4059	20732
Basic & PhD	2555	4657	1858	9070
Intermediate & Diploma	2966	1325	4756	9047
Intermediate & Bachelor's	11904	5365	18874	36143
Intermediate & Master's	8945	16285	6338	31568
Intermediate & PhD	3833	6912	2630	13375
Advanced & Diploma	2498	4652	1765	8915
Advanced & Bachelor's	9928	18842	7096	35866
Advanced & Master's	3200	28553	0	31753
Advanced & PhD	1343	12234	0	13577
Fluent & Diploma	1752	3107	1133	5992
Fluent & Bachelor's	6837	12552	4687	24076
Fluent & Master's	2057	18913	0	20970
Fluent & PhD	885	7952	0	8837
All	74617	156644	68739	300000

Table B-4 Contingency table of Language Proficiency x Education Level and Employment Status

Language Proficiency × Internship Experience	Employment Status			
	Continuing Education	Employed	Unemployed	All
Basic & Yes	11432	15227	12127	38786
Basic & No	7037	4725	9333	21095
Intermediate & Yes	17213	22778	18346	58337
Intermediate & No	10435	7109	14252	31796
Advanced & Yes	9542	44600	4499	58641
Advanced & No	7427	19681	4362	31470
Fluent & Yes	6518	29545	2883	38946
Fluent & No	5013	12979	2937	20929
All	74617	156644	68739	300000

Table B-5 Contingency table of Language Proficiency x Internship Experience and Employment Status

Education Level × Internship Experience	Employment Status			
	Continuing Education	Employed	Unemployed	All
Diploma & Yes	5719	7613	6072	19404
Diploma & No	3515	2326	4730	10571
Bachelor's & Yes	22952	30789	24331	78072
Bachelor's & No	13762	9588	18721	42071
Master's & Yes	11334	51616	5195	68145
Master's & No	8719	22957	5202	36878
PhD & Yes	4700	22132	2257	29089
PhD & No	3916	9623	2231	15770
All	74617	156644	68739	300000

Table B-6 Contingency table of Education Level x Internship Experience and Employment Status

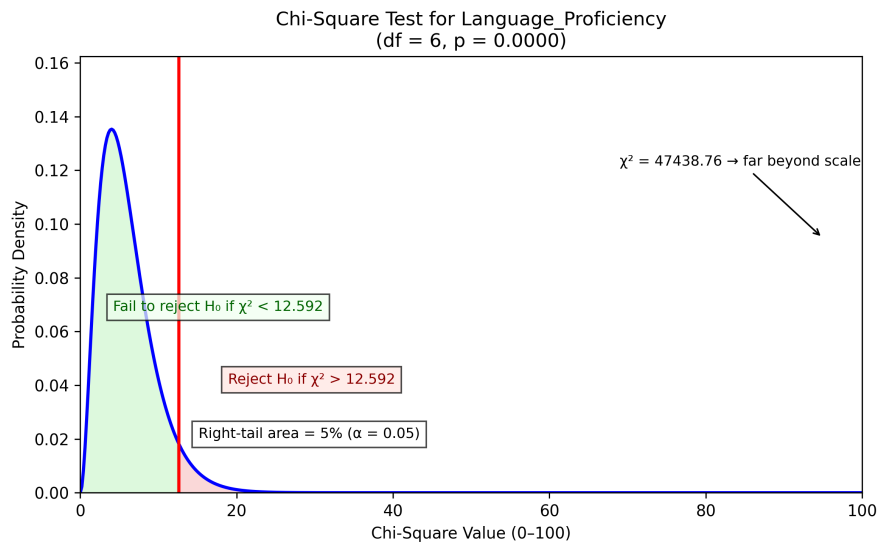


Figure B-1 Chi-Square Distribution for Language Proficiency

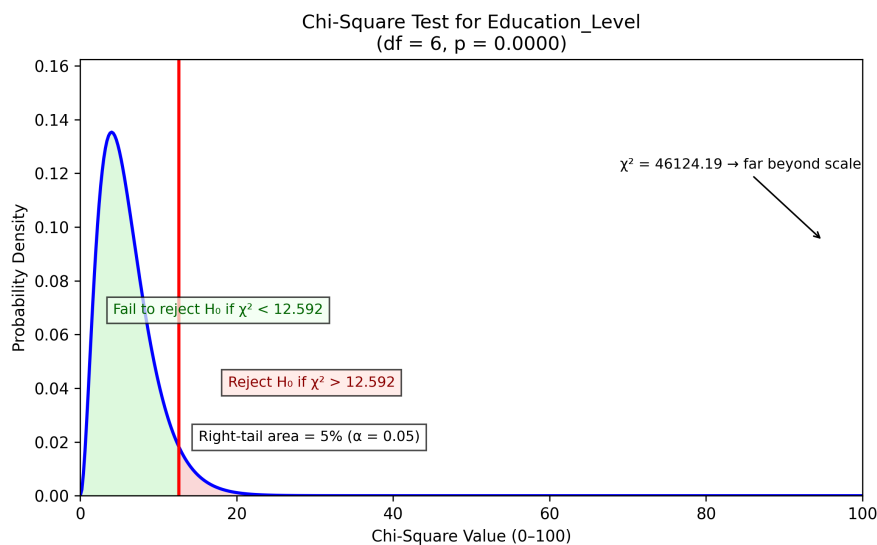


Figure B-2 Chi-Square Distribution for Education Level

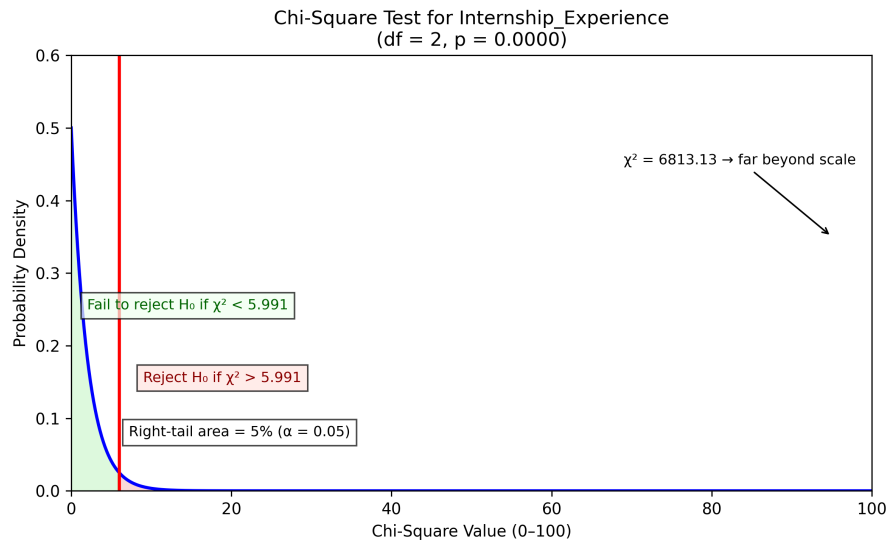
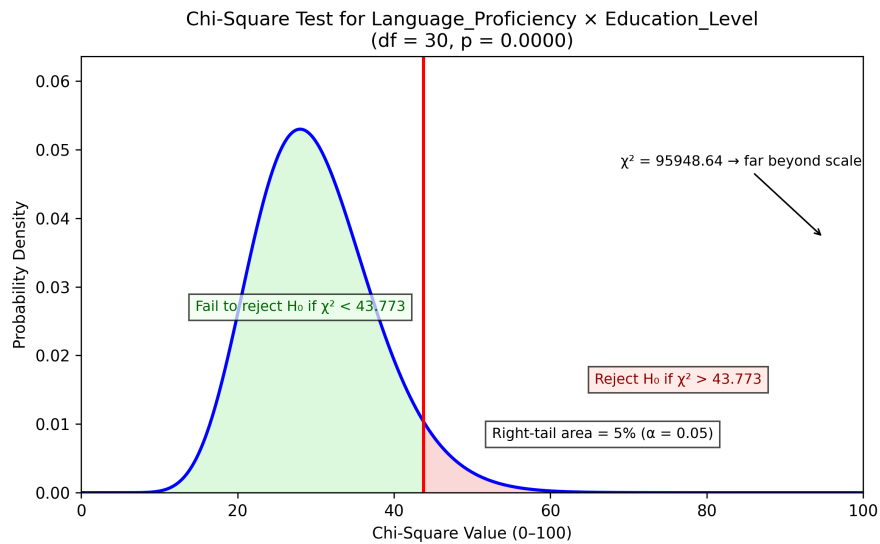


Figure B-3 Chi-Square Distribution for Internship Experience

Figure B-4 Chi-Square Distribution for Language Proficiency $\times$ Education Level

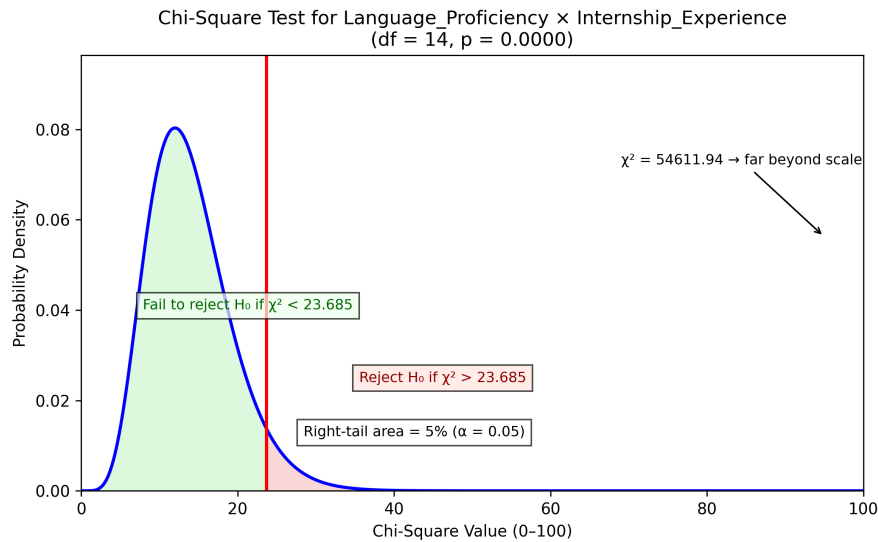


Figure B-5 Chi-Square Distribution for Language Proficiency x Internship Experience

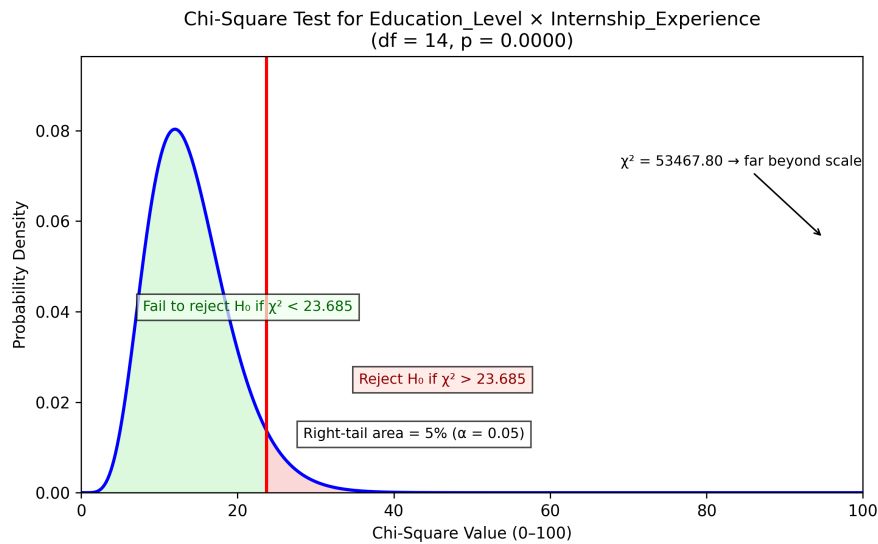


Figure B-6 Chi-Square Distribution for Education Level x Internship Experience

Appendix C

group1	group2	meandiff	p-adj	lower	upper	reject
Advanced	Basic	5338.0547	0.0	4884.7097	5791.3996	True
Advanced	Fluent	4787.3496	0.0	4438.1242	5136.575	True
Advanced	Intermediate	5347.7593	0.0	4955.7548	5739.7638	True
Basic	Fluent	-550.705	0.0169	-1030.5927	-70.8174	True
Basic	Intermediate	9.7047	1.0	-502.1559	521.5652	False
Fluent	Intermediate	560.4097	0.0037	137.9905	982.8289	True

Table C-1 Multiple Comparison of Means: Tukey HSD (Language Proficiency)

group1	group2	meandiff	p-adj	lower	upper	reject
Bachelor's	Diploma	-8961.4348	0.0	-9197.7403	-8725.1293	True
Bachelor's	Master's	16113.3618	0.0	15982.8408	16243.8829	True
Bachelor's	PhD	45124.8335	0.0	44966.44	45283.227	True
Diploma	Master's	25074.7966	0.0	24849.4374	25300.1558	True
Diploma	PhD	54086.2683	0.0	53843.6997	54328.8369	True
Master's	PhD	29011.4717	0.0	28869.9267	29153.0166	True

Table C-2 Multiple Comparison of Means: Tukey HSD (Educational Level)

Source	sum_sq	df	F	PR(>F)
C(Internship_Experience)	1.587005e+11	1.0	493.809886	3.784250e-109
Residual	3.364172e+13	104679.0	NaN	NaN

Table C-3 Internship Experience ANOVA

Appendix D

group1	group2	meandiff	p-adj	lower	upper	reject
Bachelor's-No	Bachelor's-Yes	3072.899 8	0.0	2797.4329	3348.3668	True
Bachelor's-No	Diploma-No	- 8645.694 5	0.0	-9188.6817	-8102.7073	True
Bachelor's-No	Diploma-Yes	- 5993.524 8	0.0	-6354.9042	-5632.1453	True
Bachelor's-No	Master's-No	15196.58 22	0.0	14910.174 3	15482.9901	True
Bachelor's-No	Master's-Yes	19913.73 84	0.0	19651.686 3	20175.7905	True
Bachelor's-No	PhD-No	42608.52 99	0.0	42269.232 7	42947.827	True
Bachelor's-No	PhD-Yes	49601.61 2	0.0	49313.521	49889.703	True
Bachelor's-Yes	Diploma-No	- 11718.59 44	0.0	- 12223.458 5	-11213.7303	True
Bachelor's-Yes	Diploma-Yes	- 9066.424 6	0.0	-9367.5022	-8765.347	True
Bachelor's-Yes	Master's-No	12123.68 23	0.0	11918.543 1	12328.8216	True
Bachelor's-Yes	Master's-Yes	16840.83 86	0.0	16671.357 2	17010.3199	True
Bachelor's-Yes	PhD-No	39535.63	0.0	39261.449	39809.811	True
Bachelor's-Yes	PhD-Yes	46528.71 21	0.0	46321.229 5	46736.1948	True
Diploma-No	Diploma-Yes	2652.169 8	0.0	2095.7519	3208.5876	True
Diploma-No	Master's-No	23842.27 67	0.0	23331.360 7	24353.1927	True
Diploma-No	Master's-Yes	28559.43 3	0.0	28061.761 4	29057.1046	True
Diploma-No	PhD-No	51254.22 44	0.0	50711.888 4	51796.5604	True
Diploma-No	PhD-Yes	58247.30 65	0.0	57735.445	58759.1679	True
Diploma-Yes	Master's-No	21190.10 7	0.0	20878.987 7	21501.2262	True

Diploma-Yes	Master's-Yes	25907.26 32	0.0	25618.408 6	26196.1178	True
Diploma-Yes	PhD-No	48602.05 46	0.0	48241.654 5	48962.4548	True
Diploma-Yes	PhD-Yes	55595.13 67	0.0	55282.467 4	55907.8061	True
Master's-No	Master's-Yes	4717.156 3	0.0	4530.4175	4903.8951	True
Master's-No	PhD-No	27411.94 77	0.0	27126.776 4	27697.119	True
Master's-No	PhD-Yes	34405.02 98	0.0	34183.226 9	34626.8326	True
Master's-Yes	PhD-No	22694.79 14	0.0	22434.091 5	22955.4914	True
Master's-Yes	PhD-Yes	29687.87 35	0.0	29498.563 4	29877.1837	True
PhD-No	PhD-Yes	6993.082 1	0.0	6706.2204	7279.9437	True

Table D-1 Two-way ANOVA (Educational Level $\times$ Internship Experience)

group1	group2	meandiff	p-adj	lower	upper	reject
Bachelor's-Advanced	Bachelor's-Basic	1579.323	0.0	1093.5053	2065.1406	True
Bachelor's-Advanced	Bachelor's-Fluent	3559.964 2	0.0	3251.7021	3868.2264	True
Bachelor's-Advanced	Bachelor's-Intermediate	1690.125 6	0.0	1273.2105	2107.0407	True
Bachelor's-Advanced	Diploma-Advanced	-8647.69	0.0	-9085.6544	-8209.7255	True
Bachelor's-Advanced	Diploma-Basic	- 7600.350 8	0.0	-8543.0867	-6657.6148	True
Bachelor's-Advanced	Diploma-Fluent	- 5887.793 6	0.0	-6404.3624	-5371.2247	True
Bachelor's-Advanced	Diploma-Intermediate	- 7201.704 1	0.0	-7958.7069	-6444.7013	True
Bachelor's-Advanced	Master's-Advanced	15770.16 02	0.0	15519.212 6	16021.1077	True
Bachelor's-Advanced	Master's-Basic	17090.45 38	0.0	16767.839 9	17413.0678	True

Bachelor's-Advanced	Master's-Fluent	20801.1961	0.0	20526.6286	21075.7636	True
Bachelor's-Advanced	Master's-Intermediate	17279.973	0.0	16994.1823	17565.7638	True
Bachelor's-Advanced	PhD-Advanced	43849.1589	0.0	43539.3854	44158.9324	True
Bachelor's-Advanced	PhD-Basic	46212.923	0.0	45775.4667	46650.3793	True
Bachelor's-Advanced	PhD-Fluent	51422.7105	0.0	51065.2479	51780.173	True
Bachelor's-Advanced	PhD-Intermediate	46147.4328	0.0	45770.0657	46524.7998	True

Table D-2.1 Educational Level $\times$ Language Proficiency (Tukey HSD)

group1	group2	meandiff	p-adj	lower	upper	reject
Bachelor's-Basic	Bachelor's-Fluent	1980.6413	0.0	1475.0247	2486.2579	True
Bachelor's-Basic	Bachelor's-Intermediate	110.8026	1.0	-467.506	689.1112	False
Bachelor's-Basic	Diploma-Advanced	-10227.0129	0.0	-10820.6757	-9633.3502	True
Bachelor's-Basic	Diploma-Basic	-9179.6737	0.0	-10204.0631	-8155.2843	True
Bachelor's-Basic	Diploma-Fluent	-7467.1165	0.0	-8120.9254	-6813.3077	True
Bachelor's-Basic	Diploma-Intermediate	-8781.027	0.0	-9637.5758	-7924.4783	True
Bachelor's-Basic	Master's-Advanced	14190.8372	0.0	13717.9765	14663.698	True
Bachelor's-Basic	Master's-Basic	15511.1309	0.0	14996.6385	16025.6232	True
Bachelor's-Basic	Master's-Fluent	19221.8731	0.0	18736.0646	19707.6816	True
Bachelor's-Basic	Master's-Intermediate	15700.6501	0.0	15208.4114	16192.8888	True
Bachelor's-Basic	PhD-Advanced	42269.836	0.0	41763.2965	42776.3754	True
Bachelor's-Basic	PhD-Basic	44633.6001	0.0	44040.3121	45226.8881	True
Bachelor's-Basic	PhD-Fluent	49843.3875	0.0	49306.3567	50380.418	True

Bachelor's-Basic	PhD-Intermediate	44568.10 98	0.0	44017.629 6	45118.5901	True
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Table D-2.2 Language × Education Pairwise Tukey HSD